Critical rainfall intensity of unsaturated slopes: how much do uncoupled, passive-air analyses overestimate it?

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Abstract

Physically based rainfall thresholds for unsaturated slopes are often derived from a seepage analysis with atmospheric pore air followed by a stability analysis with the computed pore-water pressures. We quantify how these hypotheses change the rainfall that triggers failure. A formulation with displacement, pore-water and pore-air pressures (Bishop's effective stress, Mohr–Coulomb plasticity) is implemented in the open-source library NeoPZ and verified against the Liakopoulos drainage test with passive air; the factor of safety (FS) is obtained by strength reduction with frozen pressures. Because the coupled transient is solved with the real strength, the minimum FS of the coupled models with an associated flow rule did not fall below one in the cases computed; models are therefore compared by the critical intensity $I_{c,\delta}$ of a 24-h rainfall, the smallest that brings the minimum FS to $1+\delta$, and by the time to failure. For the slope of Rahardjo et al. (2007) with $k_s = 36$ mm/h and $\delta = 0.005$–0.02, the uncoupled, passive-air model gives $I_{c,\delta} = 47$–49 mm/h ($1.3k_s$); the coupled model with active air gives 30–35 mm/h with an associated flow rule and 26 mm/h without dilatancy. The usual hypotheses thus overestimate the failure-triggering rainfall by 40–59% and 85–88%, respectively, and delay the predicted failure by 4.5 and 9.9 h with 54 mm/h. Trapped air halves $I_{c,\delta}$; coupling raises it, slightly without dilatancy and strongly with an associated flow rule. With active air less water infiltrates and the matric suction is higher, but the pore-water pressure rises with the air pressure; a low-permeability slope that the usual model keeps 9–10% above failure comes within 0.009–0.015 of it. Without dilatancy the coupled transient collapses; with $\psi \geq \phi'/2$ it does not. The bias persists with $\chi = S_e$ and a finer mesh, and changes sign in a permeable soil ($k_s = 360$ mm/h). In these analyses the pore air is far from atmospheric once the surface ponds (36–40 kPa), which questions the assumption that it is approximately atmospheric in most in situ conditions, and the adequacy of passive-air and uncoupled analyses depends on the process and on the flow rule, contrary to general statements in the literature.

Keywords: unsaturated slope, rainfall infiltration, pore-air pressure, hydro-mechanical coupling, dilatancy, critical rainfall intensity, strength reduction method

1. Introduction

Intensity–duration thresholds for shallow landslides, $I = aD^b$ with a and b obtained by statistical analyses, were first proposed by Caine (1980) and are used at local, regional and global scales (Guzzetti et al., 2008), as summarized by Wang et al. (2021, p. 2913). Physically based thresholds come instead from analyses of the slope. In geotechnical practice

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the stability of slopes under rainfall is "often performed using a two-stage method", a seepage analysis followed by a stability analysis with the computed pore-water pressures, or with a finite element method with hydro-mechanical coupling (Liu and Wang, 2021, p. 1); "it is common practice to import data from seepage analyses into stability codes" (Pedroso, 2015, p. 791). Rahardjo et al. (2007, pp. 1534–1535) and Tang et al. (2018, p. 324) solved the transient seepage with SEEP/W and computed the factor of safety (FS) with SLOPE/W; Pradhan et al. (2022, p. 4) used a fully coupled flow–deformation analysis in PLAXIS 2D, based on Biot's theory, and the strength reduction method.

The parametric study of Rahardjo et al. (2007) on a homogeneous slope is a reference of the uncoupled approach. They concluded that the initial water table and the slope geometry "only played a secondary role" and that "for a given rainfall duration, there was a threshold rainfall intensity which would produce the global minimum factor of safety" (p. 1532); this threshold intensity $I_{r(\mathrm{thr})}$, defined at the lower inflection point of a sigmoid fitted to the minimum FS, increases with the saturated permeability k_s (p. 1539); and slopes with $k_s \leq 10^{-6}$ m/s, even at 63°, "will not fail under short-duration rainfall ($T_r \leq 24$ h) even when the rainfall intensity is extremely high" (p. 1541), because most of the rainwater runs off (p. 1539). Three hypotheses of this approach are examined here.

1. *Atmospheric pore air.* Sheng et al. (2008, p. 811) take the air pressure at the atmospheric value as "approximately true for most in situ conditions"; Li et al. (1999, p. 136) and Pedroso (2015, p. 792) adopt it to simplify the model, and the coupled slope analyses of Wang et al. (2018, p. 3), Liu and Wang (2021, p. 4) and Pradhan et al. (2022, p. 4) also neglect the air pressure. Fredlund (2006, p. 292) restricts the assumption to "situations where the soil is exposed to atmospheric pressure and the air phase is continuous". Comparisons of passive-air and fully coupled analyses "are rare" (Ghorbani et al., 2018, p. 4), and the literature disagrees on their outcome: Khoei and Saeedmonir (2021, p. 2) state that the passive-gas assumption leads "to unrealistic outcomes", citing among others Callari and Abati (2009), whose passive-air solution of the Liakopoulos drainage test is "in good agreement with available experimental data" (Callari and Abati, 2009, p. 491).
2. *Uncoupled seepage.* Pradhan et al. (2022, p. 2) report that coupled analyses "have been reported to provide a better prediction of factor of safety"; Fredlund (2006, p. 315) notes that "the uncoupled simultaneous solution yields satisfactory results for geotechnical engineering purposes". Neither quantifies the effect of the coupling on the rainfall that triggers a slope failure.
3. *Flow rule.* The coupled analyses of rainfall-induced slope failure cited above use a dilation angle of 0.1° (Wang et al., 2018, p. 6) or zero (Liu and Wang, 2021, p. 5); Pradhan et al. (2022, Table 2) do not list one. None of the studies cited here varies the dilatancy.

These studies compare analyses by the minimum FS during and after the rainfall (e.g. Tang et al., 2018, p. 324).

This paper quantifies how these three hypotheses change the critical rainfall intensity of an unsaturated slope, the smallest intensity of a rainfall of given duration that brings the slope to failure. A three-field formulation (displacement, pore-water and pore-air pressures) is implemented in the open-source finite element library NeoPZ (Devloo, 1997) and verified

against the drainage test of Liakopoulos (1965), with the measurements reported by Callari and Abati (2009). The slope of Rahardjo et al. (2007) is the reference case: four combinations of coupling and air model separate the contribution of each hypothesis, and the effect of the dilatancy, of the soil permeability, of the Bishop parameter, of the confinement of the air and of the mesh is then assessed. Section 9 confronts the results with the statements of the literature listed above.

2. Formulation and implementation

2.1. Governing equations

The soil is a deformable porous medium with two immiscible fluids, water and air, under isothermal conditions; grains and water are nearly incompressible and the air is an ideal gas; dissolved air and vapour are neglected. Small strains and quasi-static equilibrium are assumed. With tension positive and pore pressures positive in compression, the total stress is

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}' - p_B\,\mathbf{I}, \qquad p_B = \chi\,p_w + (1-\chi)\,p_a, \tag{1}$$

where $\boldsymbol{\sigma}'$ is Bishop's effective stress (Bishop, 1959) and χ the Bishop parameter. The unknowns are the displacement $\mathbf{u}$ and the gauge pressures p_w and p_a; the capillary pressure is $p_c = p_a - p_w$. The balance equations of momentum, water mass and air mass are

$$\nabla\cdot\boldsymbol{\sigma} - \gamma\,\mathbf{e}_y = \mathbf{0}, \tag{2}$$

$$\frac{nS_w}{K_w}\,\dot{p}_w + n\,\dot{S}_w + S_w\,\dot{\varepsilon}_v + \nabla\cdot\mathbf{q}_w = 0, \tag{3}$$

$$\frac{nS_a}{p_a+p_{atm}}\,\dot{p}_a - n\,\dot{S}_w + S_a\,\dot{\varepsilon}_v + \nabla\cdot\mathbf{q}_a + \frac{\mathbf{q}_a\cdot\nabla p_a}{p_a+p_{atm}} = 0, \tag{4}$$

with $S_a = 1 - S_w$, $S_w = S_w(p_c)$, ε_v the volumetric strain (positive in expansion), γ the total unit weight, $\mathbf{e}_y$ the upward unit vector, a unit Biot coefficient and Darcy's law

$$\mathbf{q}_w = -\frac{k_{rw}}{\gamma_w}\,\mathbf{K}\,(\nabla p_w + \gamma_w\mathbf{e}_y)\,, \qquad \mathbf{q}_a = -\frac{k_{ra}}{\gamma_w}\,\frac{\mu_w}{\mu_a}\,\mathbf{K}\,(\nabla p_a + \gamma_a\mathbf{e}_y)\,, \tag{5}$$

where $\mathbf{K}$ is the saturated conductivity tensor, k_{rw} and k_{ra} the relative permeabilities and $\mu_w/\mu_a = 55.6$. The air mass balance is the volume balance of the mass equation divided by the air density $\rho_a \propto p_a + p_{atm}$: its flux term is $\rho_a^{-1}\nabla\cdot(\rho_a\mathbf{q}_a) = \nabla\cdot\mathbf{q}_a + \mathbf{q}_a\cdot\nabla p_a/(p_a+p_{atm})$. The last term is the term $(\dot{\mathbf{w}}^g/\rho_g)\cdot\nabla\rho_g$ of Ghorbani et al. (2018, Eq. 35) and the term $\dot{w}_{gi}\,\partial\rho_g/\partial x_i/(S_g\rho_g)$ of Li et al. (1999, Eq. 36), both obtained from a mass balance with the gas density inside the divergence (Li et al., 1999, Eq. 33).

As in Rahardjo et al. (2007, pp. 1533–1534), the retention curve is that of Fredlund and Xing (1994) with the correction factor $C(\psi) = 1$, $S_w = [\ln(e + (p_c/a)^n)]^{-m}$, and the water permeability is $k_w = k_s\Theta^p$ (Leong and Rahardjo, 1997) with $\Theta = \theta_w/\theta_s = S_w$ and $p = 4$. The air relative permeability is the Brooks–Corey form of Khoei and Saeedmonir (2021, Eq. 48), $k_{ra} = (1-S_e)^2(1-S_e^{(2+\lambda)/\lambda})$ with $\lambda = 3$ and $S_e = S_w$, bounded below by $k_{ra}^{\min} = 10^{-4}$.

The effective stress follows the Mohr–Coulomb criterion with elastic moduli E, ν. With an associated flow rule ($\psi = \phi'$) the return mapping is analytical, with the consistent tangent;

with $\psi < \phi'$ an iterative non-associated return mapping is used. The strength envelope $\tau = c' + (\sigma_n - u_a)\tan\phi' + (u_a - u_w)\tan\phi^b$ (Fredlund et al., 1978), used by Tang et al. (2018, p. 324), corresponds to $\chi = \tan\phi^b/\tan\phi'$ on the suction; Rahardjo et al. (2007, p. 1533) use $\phi^b = \phi' = 26°$, i.e. $\chi = 1$, which is the default here. Fredlund (2006, p. 312) reports that the shear strength increases with ϕ' "for matric suctions up to the air entry value of the soil" and at a decreasing rate beyond it; $\chi = S_e$ is therefore examined in Section 8.

Four models are compared (Table 1). In the uncoupled models the terms $S_w\dot{\varepsilon}_v$ and $S_a\dot{\varepsilon}_v$ are removed from Eq. (4), i.e. the flow is solved on a rigid skeleton. In the passive-air models $p_a = 0$ in the whole domain and the air mass balance is not solved. Model (iii) has the hypotheses of the analyses of Rahardjo et al. (2007) and Tang et al. (2018): seepage on a rigid skeleton (SEEP/W) and the stability computed with the resulting pore-water pressures.

Table 1: Models compared.

model	coupling	air
(i) complete	coupled	active ($p_a = 0$ only on the ground surface)
(ii)	coupled	passive ($p_a = 0$)
(iii) the usual one	uncoupled	passive
(iv)	uncoupled	active (two-phase flow on a rigid skeleton)

2.2. Boundary conditions

The rainfall is a flux on the horizontal projection of the ground surface with the unilateral condition $p_w \leq 0$, inflow $\leq I_r$ and complementarity, solved as an active set of ponded vertices; without rainfall the same condition is a seepage face. Conditions of this type are discussed by Pedroso (2015, Section 3), who notes their analogy with the complementarity conditions of plasticity. Rahardjo et al. (2007, p. 1534) applied "a boundary flux, q, equal to the desired rainfall intensity" to the surface of the slope. The pore air is atmospheric on the ground surface and the base and lateral boundaries are closed to the air (the effect of opening them is examined in Section 6.1). As in Rahardjo et al. (2007, p. 1534), the water pressure is hydrostatic on the lateral boundaries below the water table and the lateral boundaries are impervious above it. The base is fixed and the lateral boundaries are on rollers.

2.3. Discretization and solution

Displacements are interpolated by quadratic and pressures by linear triangles (Taylor–Hood), in plane strain. The storage terms are lumped at the vertices: the change of saturation as $n(S_w^{k+1} - S_w^k)$ and the compressibility of the air as $nS_a^{k+1}(p_a^{k+1} - p_a^k)/(p_a^{k+1} + p_{atm})$, where k is the time level. The equations are integrated by the implicit Euler method and solved by Newton's method with the complete Jacobian, including the derivatives of the permeabilities, of the saturation and of χ. Bergamaschi and Putti (1999, p. 1032) relate the stagnation of the Picard method in Richards' equation to "the absence of the derivative terms in the Jacobian matrix". Dirichlet conditions are eliminated and the linear systems are solved with Pardiso. A time step that does not converge is restarted from its initial state with up to five levels of bisection. The time step is 15 min during the rainfall, 1 h in the following 24 h and 4 h afterwards.

The global mass balances of water and air were checked in model (iv) with 36 mm/h, with the volume of each fluid obtained from the lumped vertex values and the fluxes through the ground surface and the boundaries with prescribed pressure. [balanço: erro de água e de ar após 24 e 48 h, a preencher com scripts/balanco_massa.py]

2.4. Factor of safety

At selected instants an independent stability analysis is made with p_w and p_a frozen: from the geostatic state with the strength increased (c'/F_0, $\tan\phi'/F_0$ with $F_0 = 0.6$) the strength reduction factor F is increased until the equilibrium no longer converges; the FS is the last converged F, "the ratio of the shear strength parameters of soil to those at failure" (Pradhan et al., 2022, p. 4), as in Griffiths and Lane (1999). It is found by bisection with a tolerance of 0.002 in the cases near the critical intensity and 0.01 in the others. The state of the transient is restored afterwards. The strength reduction is always made with $\psi = \phi'$, also for the transients computed with $\psi < \phi'$.

The transient is solved with the real strength in the coupled models: while it converges, a state in equilibrium with the real strength and with the pressures of that instant exists. In all the coupled cases with $\psi = \phi'$ computed here, the strength reduction, which starts from the same pressures, gave a minimum FS of 1.000 or more (Section 5). The loss of equilibrium with the real strength (the transient does not converge even with five bisections) is recorded as the time t_f; the transient then continues with the strength increased (F_0), only to follow the pore pressures. In the uncoupled models the deformation does not change the pressures, the transient mechanics is solved with the strength increased ($F = 0.25$) and the FS of the strength reduction can fall below one.

3. Verification

3.1. Drainage of a sand column (Liakopoulos test)

In the test of Liakopoulos (1965) a 1-m column of Del Monte sand, initially saturated with $p_w = 0$, drains through its base. The test has been "considered by several authors to validate their numerical formulations" (Callari and Abati, 2009, p. 490). The parameters are those of Callari and Abati (2009, Table 1 and Eq. 58): $E = 1.3$ MPa, $\nu = 0.4$, $n = 0.2975$, permeability 4.5×10^{-7} m^2/(kPa s) ($k_s = 4.41 \times 10^{-6}$ m/s), $S_w = 1 - 1.9722 \times 10^{-11} p_c^{2.4279}$ (p_c in Pa) and $k_{rw} = 1 - 2.207(1 - S_w)^{1.0121}$; $\chi = S_w$, elastic soil. For the air, which Callari and Abati (2009, p. 490) do not model ("A constant atmospheric value is assumed for the gas pressure"), we assume the Brooks–Corey permeability with $\lambda = 3$ and a residual saturation of 0.2. The air pressure is atmospheric at both ends, the water pressure at the base, and the top is impervious to water. The column was discretized with 2×80 cells of two triangles and time steps of 30 s.

The passive-air solution reproduces the measurements (Fig. 1c,d): the outflow differs from the measured one by 0.29×10^{-6} m/s in root mean square (maximum 0.97×10^{-6} m/s, in the first minutes) and the water pressures by 0.37–0.52 kPa in root mean square at 600, 1200 and 1800 s. The two-phase solution, with the assumed air permeability, drains more slowly than the test: its outflow is 2.0–2.7 times smaller than that of the passive-air solution between 10 and 30 min and the water pressure at 600 s differs from the measurements by 2.0 kPa in root mean square. The test therefore verifies the implementation with passive air; it does

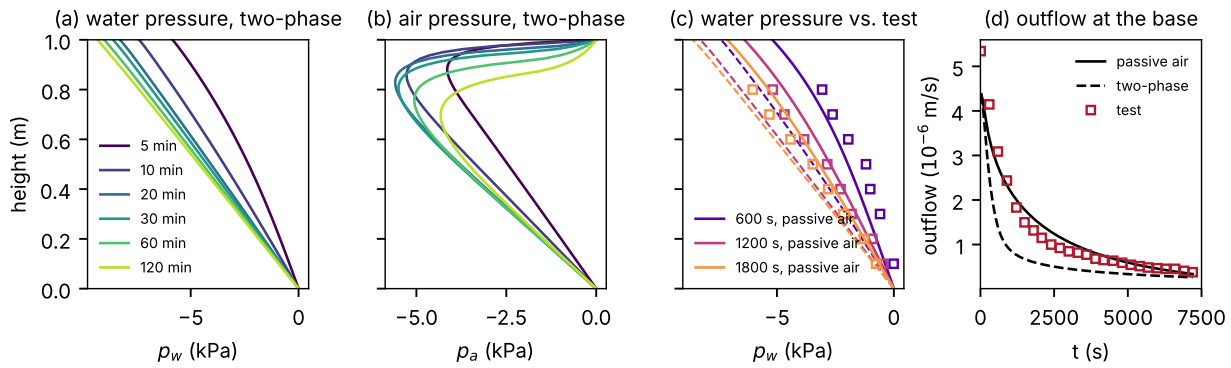


Figure 1: Drainage of the Liakopoulos column. (a) Water and (b) air pressure profiles of the two-phase model. (c) Water pressure of the passive-air model (solid) and of the two-phase model (dashed) and the measured values (squares, digitized from Callari and Abati, 2009, Fig. 3b). (d) Outflow at the base and the measured outflow (digitized from Callari and Abati, 2009, Fig. 3a).

not validate the air phase, whose effect in drainage depends on an air permeability that the test does not measure. This agrees with Callari and Abati (2009, p. 490), for whom the zero gas pressure "is acceptable in many situations of interest and avoids problems encountered with three-phase formulations in simulating the transition from full to partial saturation".

3.2. Reference slope

The slope of Rahardjo et al. (2007) is 10 m high, inclined at 45°, with 30-m platforms at the crest and at the toe and the base 20 m below the toe; the initial water table is 5 m below the toe and inclined at 7°, and the initial pore-water pressure is hydrostatic "with a limiting pore-water pressure of −75 kPa" (Rahardjo et al., 2007, p. 1535) (Table 2, Fig. 2). The standard mesh has 1684 triangles, with elements of 0.75 m down to 2 m below the surface.

Table 2: Parameters of the reference slope (Rahardjo et al., 2007, pp. 1533–1534, Table 1); E, ν, ψ and the air parameters are assumptions of this study.

soils f_{10}, f_{50}, f_{100}	a = 10, 50, 100 kPa; $k_s = 10^{-4}$, 10^{-5}, 10^{-6} m/s (360, 36 and 3.6 mm/h)
retention, k_w	Fredlund–Xing, $C(\psi) = 1$, $n = m = 1$, $\theta_s = 0.45$; $k_w = k_s\Theta^4$
strength	$c' = 10$ kPa, $\phi' = 26°$, $\phi^b = 26°$ ($\chi = 1$), $\gamma = 20$ kN/m^3
elasticity, dilatancy	$E = 20$ MPa, $\nu = 0.3$; $\psi = \phi'$ (associated), $\phi'/2$ or 0
air	Brooks–Corey $\lambda = 3$, $k_{ra}^{\min} = 10^{-4}$, $\mu_w/\mu_a = 55.6$
initial state	water table 5 m below the toe, inclined at 7°; suction ≤ 75 kPa
rainfall	uniform, 24 h; 48 h simulated

The initial FS is 2.119; the regression of Rahardjo et al. (2007, Table 7) gives 2.118 at 45°, and for heights of 5 to 40 m and water tables 2.5 to 15 m deep our initial FS differs from their Figs. 8 and 10 by at most 0.05. A simplified Bishop analysis with a search of circular surfaces on the same initial pressures gives 2.092. At 63.4° the initial FS is 1.775, against

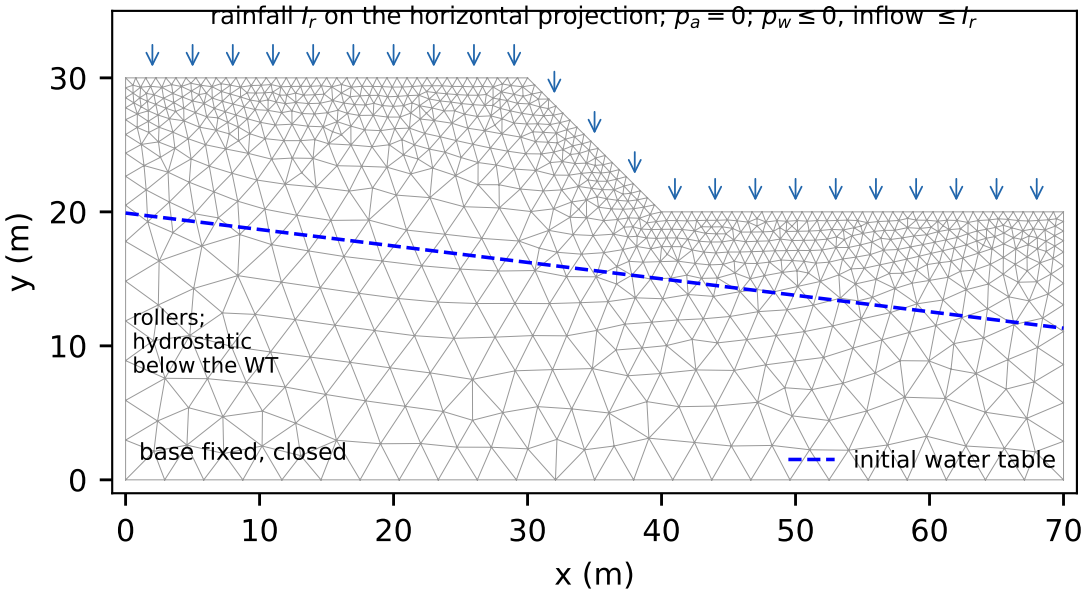


Figure 2: Reference slope (Rahardjo et al., 2007): geometry, standard mesh, initial water table and boundary conditions.

1.695 from the same regression. With model (iii) the minimum FS under 24-h rainfalls of the three soils is 0.01–0.14 above the sigmoid fitted by Rahardjo et al. (2007, Eq. 6 and Fig. 7), never below it; the largest differences are in the transition between the upper and the lower plateau. The comparison uses their figures: five of the ten regression lines of their Table 7 do not reproduce the lines drawn in their Fig. 9 (e.g. f_{100} with 80 mm/h: $1.455 - 0.013\alpha$ in the table, i.e. 0.87 at 45°, against about 1.46 in Fig. 9c and the lower asymptote $A = 1.45$ of Fig. 7c), and their text gives $I_{r(\mathrm{ini})} = 3.8$ mm/h for the f_{100} soil where Fig. 7c gives 2.8 mm/h ($0.778k_s$). Of the eight laws for the suction strength compared by Zhang et al. (2014, Tables 4 and 5), the three linear ones ($\phi^b = 0$, 15° and ϕ') are reproduced in their steep and flat slopes within 0.05–3.9%.

4. Failure criterion and critical intensity

The slope fails when the FS reaches one. The FS of each instant is computed with the pressures of that instant frozen; a value below one indicates that the available strength is insufficient for the analysed configuration and that equilibrium with that strength is not possible. It does not describe the subsequent deformation or the redistribution of the pore pressures, which the frozen-pressure analysis does not compute. The quantity compared is therefore whether, with which rainfall and at which instant, the FS reaches one.

Because the minimum FS of the coupled models with an associated flow rule did not fall below one in any case computed (Section 2.4), the critical intensity is defined with a threshold $1 + \delta$: $I_{c,\delta}$ is the smallest intensity of a 24-h rainfall whose minimum FS is $\leq 1 + \delta$, obtained by linear interpolation between the computed intensities that bracket it, and the time to failure is the first instant with FS $\leq 1 + \delta$ (or t_f, if the coupled transient loses equilibrium earlier). The values are given for $\delta = 0$, 0.005, 0.01 and 0.02; δ must exceed the bisection

tolerance (0.002) to be meaningful in the coupled associated models, and the mesh error of the minimum FS is 0.003–0.023 (Section 6.2). Unless stated otherwise, I_c denotes $I_{c,0.01}$.

Rahardjo et al. (2007, p. 1539) call threshold rainfall intensity $I_{r(\mathrm{thr})}$ the intensity beyond which the minimum FS "remains constant"; it is not the intensity that brings the FS to one, and for the soil f_{50} it is 51 mm/h ($1.42k_s$, their Fig. 7b). The lower FS plateau of the intense rainfalls is, in model (iii), the FS of the slope saturated up to its surface with the water flowing through it. A steady-state calculation with the surface ponded gives 0.84–0.85 for the three soils; in the transient with the surface ponded it is reached between 6 and 12 h, 12 and 24 h, and 96 and 192 h of ponding in the f_{10}, f_{50} and f_{100} soils (the instants computed). It depends on the orientation of an anisotropic conductivity tensor (0.63–1.05 for $k_1/k_2 = 10$), not on the magnitude of k_s. This agrees with Li et al. (2024, p. 9), for whom the FS of a slope saturated by a steady infiltration "is identical to that without considering suction", and with Cheng et al. (2024, p. 17), who attribute the constant minimum FS above a certain intensity to the degree of saturation, which "keeps constant when the rainfall intensity exceeds a certain value". The lower asymptotes of Rahardjo et al. (2007, Fig. 7) are 0.81 and 0.82 for the f_{10} and f_{50} soils and 1.45 for the f_{100} soil; with 24-h rainfalls the f_{100} slope does not reach the saturated state in model (iii) either, so its plateau is not the FS of the saturated slope. These statements concern model (iii) and the reference slope.

5. Critical intensity of the reference slope

Figure 3 and Table 3 give the minimum FS under 24-h rainfalls of the f_{50} soil (k_s = 36 mm/h) in the four models, the critical intensities $I_{c,\delta}$ and the times to failure. The usual model (iii) gives $I_{c,\delta}$ = 47–49 mm/h ($1.3k_s$) for δ = 0–0.02. The complete model (i) gives 30, 31 and 35 mm/h with $\psi = \phi'$ for δ = 0.02, 0.01 and 0.005, and 26 mm/h with $\psi = 0$ for all δ; with $\psi = \phi'$ and $\delta = 0$ it has no critical intensity, its minimum FS being 1.000–1.003 from 36 to 180 mm/h. For δ = 0.005–0.02 the hypotheses of the usual chain therefore overestimate the rainfall that brings the slope to failure by 40–59% with $\psi = \phi'$ and by 85–88% with $\psi = 0$. The sigmoid of Rahardjo et al. (2007, Fig. 7b) reaches $1 + \delta$ at 38–40 mm/h: model (iii) is 23–24% above it, model (i) 12–22% below it with $\psi = \phi'$ and 33–34% below it with $\psi = 0$. The usual hypotheses also delay the failure: with 54 mm/h the FS of model (iii) reaches 1.01 after 22.4 h, against 17.9 h in model (i) with $\psi = \phi'$ and the loss of equilibrium after 12.5 h with $\psi = 0$; with 90 mm/h, after 19.7, 18.3 and 12.0 h. In these cases the FS was computed every 3 h after the third hour, and the times are interpolated between the outputs.

The four models separate the two hypotheses (Table 3).

- *The air lowers the critical intensity by a factor of 2.0–2.5*, for the same coupling and flow rule: on a rigid skeleton from 47–49 to 24–25 mm/h (models iii and iv, δ = 0–0.02); in the coupled models from at most 52–53 to 26 mm/h without dilatancy (models ii and i; the run of model ii with 54 mm/h stopped at 24 h, after its loss of equilibrium at 22.3 h) and from 62 to 30 mm/h (δ = 0.02) and from 78 to 31 mm/h (δ = 0.01) with $\psi = \phi'$.

- *The coupling raises it*: without dilatancy by 5–7% with active air (from 24–25 to 26 mm/h) and by at most 11% with passive air (from 47–48 to at most 52–53 mm/h);

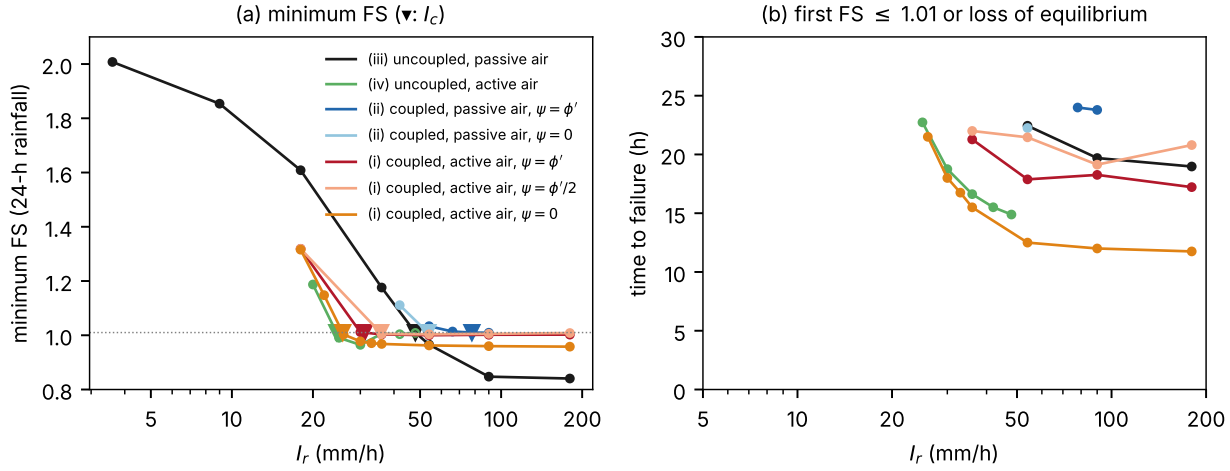


Figure 3: Soil f_{50}, $\alpha = 45°$, 24-h rainfalls. (a) Minimum FS against the rainfall intensity in the four models (▼: $I_{c,0.01}$; open symbols: runs stopped at the first FS ≤ 1.01, whose minimum FS is lower). (b) Time to failure: first instant with FS ≤ 1.01, or loss of equilibrium of the coupled transient with $\psi = 0$.

Table 3: Critical intensity $I_{c,\delta}$ ($k_s = 36$ mm/h) and time to failure of the reference slope. $>$: no failure up to the most intense rainfall computed; $\leq$: upper bound; n.c.: not computable with the runs available (run stopped at the first FS ≤ 1.01). Bracket: computed intensities on each side of $I_{c,0.01}$. Lowest min. FS: smallest minimum FS of the runs that were not stopped. Time to failure: first instant with FS ≤ 1.01; –: no failure; blank: not computed; †loss of equilibrium of the coupled transient. With $\psi = \phi'/2$ only 18 and 36 mm/h were computed below 54 mm/h.

model	$I_{c,\delta}$ (mm/h) for $\delta =$				bracket	I_c/k_s	lowest	time to failure (h), $I_r =$		
	0.02	0.01	0.005	0	($\delta = 0.01$)		min. FS	36	54	90
(iii) uncoupled, passive air	47	48	49	49	48–54	1.33	0.841	–	22.4	19.7
(iv) uncoupled, active air	24	25	25	25	20–25	0.68	0.965	16.6		
(ii) coupled, passive, $\psi = \phi'$	62	78	> 90	> 90	66–78	2.16	1.009		–	23.8
(ii) coupled, passive, $\psi = 0$	≤ 52	≤ 53	n.c.	n.c.	42–54	1.49	1.111		22.2†	
(i) coupled, active, $\psi = \phi'$	30	31	35	> 180	30–36	0.86	1.000	21.3	17.9	18.3
(i) coupled, active, $\psi = \phi'/2$	35	36	36	> 180	18–36	0.99	1.004	22.0	21.5	19.1
(i) coupled, active, $\psi = 0$	26	26	26	26	22–26	0.72	0.958	15.5†	12.5†	12.0†

with $\psi = \phi'$ by 22–41% with active air and by 32% ($\delta = 0.02$) to 62% ($\delta = 0.01$) with passive air; with $\delta = 0.005$ the critical intensity of model (ii) exceeds 90 mm/h. Models (iii) and (ii), and models (iv) and (i), differ only by the volumetric strain terms of Eq. (4); the mechanism of the increase (elastic or plastic volume change) was not separated.

The two simplifications of the usual model thus have opposite effects, and the effect of the air is the larger one. A two-phase flow analysis on a rigid skeleton, model (iv), gives a critical intensity 5–6% below that of the complete model without dilatancy and 18–29% below that with $\psi = \phi'$, and its time to failure with 36 mm/h, 16.6 h, lies between those of the complete model with $\psi = 0$ (15.5 h) and with $\psi = \phi'$ (21.3 h).

The coupled models with an associated flow rule approach one from above (1.000–1.014 for $I_r \geq I_{c,0.02}$), so their critical intensity depends on δ: from $\delta = 0.02$ to 0.005 that of model (i) rises from 30 to 35 mm/h and that of model (ii) from 62 to more than 90 mm/h, while those of the uncoupled models and of the models without dilatancy change by less than 1.5 mm/h. In model (ii) with $\psi = \phi'$ the minimum FS is 1.009–1.014 from 66 to 90 mm/h: the slope is in limit equilibrium, within the resolution of the FS, over a wide range of intensities, with plastic deformation but without loss of equilibrium.

6. Trapped air

Rahardjo et al. (2007, pp. 1539, 1541) conclude that slopes with $k_s \leq 10^{-6}$ m/s, "even when the rainfall intensity is extremely high", do not fail under rainfalls of up to 24 h, because the rain does not raise the water table and most of it runs off. We examine the f_{100} soil ($k_s = 3.6\,\text{mm/h}$) on the steepest slope of their study, $\alpha = 63.4°$, whose initial FS is 1.775 in our analysis (Fig. 4); their Fig. 9c gives about 1.69 for the initial FS and about 1.12 and 1.05 for the minimum FS with 9 and 80 mm/h (read from the figure). With the usual model (iii) the minimum FS under 24-h rainfalls is 1.383, 1.163, 1.097 and 1.093 for $I_r = 9$, 18, 36 and 80 mm/h: the slope does not fail, as Rahardjo et al. (2007) conclude. With model (ii) the minimum FS is 0.015–0.041 higher. With active air (model i) it is 1.166, 1.028, 1.015 and 1.013: from 36 mm/h on, the minimum FS is within 0.02 of one at the end of the rainfall, i.e. the slope fails with $\delta = 0.02$ ($I_{c,0.02} = 29$ mm/h) but not with $\delta = 0.01$. With elements of 0.35 m near the surface the minimum FS with 36 mm/h is 1.009 (Section 6.2).

The lower FS of model (i) is not associated with more water. With active air 10.4–11.6 m^3/m infiltrate in 24 h, against 13.5–18.0 m^3/m with passive air (Fig. 4b). The surface ponds earlier with active air: with 9 mm/h the first vertex ponds after 6 h with active air and after 11.75 h with passive air, and at the end of the rainfall 83 and 41 of the 96 surface vertices are ponded; with 36 mm/h the whole surface ponds within 4.5 h with active air and 6.5 h with passive air. With 36 mm/h the maximum air pressure is 19.5 kPa after 1 h, 30 kPa after 3 h and 38 kPa after 24 h. At the end of the rainfall, along a vertical below the middle of the face (Fig. 4c,d), the air pressure averages 32 kPa between 0.5 and 5 m of depth; between 2 and 5 m the matric suction is 38 kPa with active air and 24 kPa with passive air, and the water pressure is $-6\,\text{kPa}$ with active air and $-24\,\text{kPa}$ with passive air. The matric suction is therefore higher with active air, while the water pressure, which with $\chi = 1$ is the only pressure in the effective stress (Eq. 1), is also higher. Across the models,

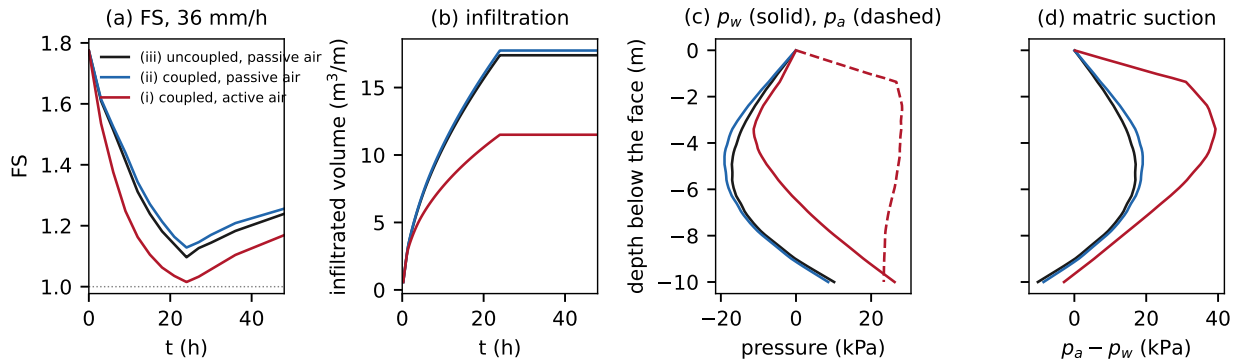


Figure 4: Soil f_{100}, $\alpha = 63.4°$, $I_r = 36\,\mathrm{mm/h}$ during 24 h, in models (i)–(iii). (a) FS; (b) infiltrated volume; (c) water (solid) and air (dashed, model i) pressures and (d) matric suction along a vertical below the middle of the face at $t = 24$ h.

a smaller infiltrated volume comes with a smaller FS. Rahardjo et al. (2007, p. 1539) relate the decrease of the minimum FS to the rainfall volume needed to initiate and to complete it; within each of our models the minimum FS also decreases with the intensity, but between models with and without active air the ordering of volume and FS is reversed.

The air pressure depends on how easily the air leaves through the ponded surface. With the minimum relative permeability of the air $k_{ra}^{\min}$ between 10^{-6} and 10^{-3} the minimum FS is 1.013–1.015 and the maximum air pressure 37–38 kPa; with 10^{-2} the air pressure falls to 28 kPa and the minimum FS rises to 1.046.

6.1. Confinement of the air

The base and the lateral boundaries are closed to the air in the reference model. Opening them (atmospheric air pressure on the base and on the lateral boundaries) changes the minimum FS by less than 0.002: 1.017 against 1.015 for the f_{100} slope with 36 mm/h, and 1.011 against 1.011 for the f_{50} slope with 30 mm/h. The lateral boundaries are 30 m away from the slope and the base is below the water table.

6.2. Mesh

The minimum FS of the f_{100} slope with 36 mm/h decreases with the mesh: 1.0154, 1.0122 and 1.0090 with elements of 0.75, 0.5 and 0.35 m near the surface (1620, 3597 and 7314 triangles), by about 0.003 per refinement, while the initial FS changes less at each refinement (1.775, 1.761 and 1.752; Fig. 5). Whether the converged minimum FS falls below one cannot be decided with these meshes.

7. Dilatancy and the response after failure

The dilatancy changes the critical intensity and, above all, what the coupled transient does after it (Figs. 6 and 3). Without dilatancy ($\psi = 0$) the coupled transient of model (i) loses equilibrium for $I_r \geq 26\,\mathrm{mm/h}$, between 21.5 h (26 mm/h) and 11.75 h (180 mm/h); with 22 mm/h it does not. With $\psi = \phi'/2$ or $\psi = \phi'$ the minimum FS stays between 1.000 and 1.008 from 36 to 180 mm/h and the transient never loses equilibrium: the slope is at

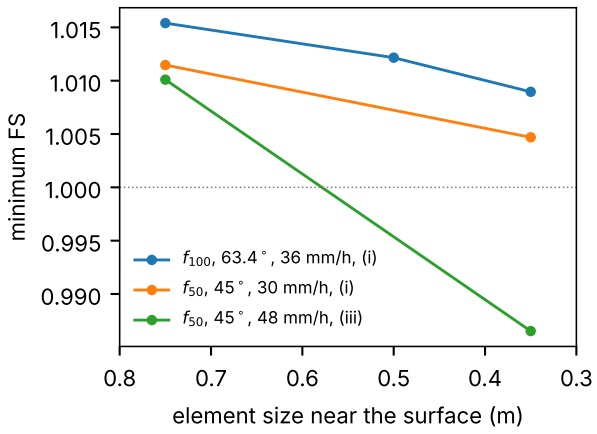


Figure 5: Minimum FS against the element size near the surface: f_{100} slope at 63.4° with 36 mm/h in model (i), and the f_{50} reference slope near its critical intensities in models (i) and (iii).

limit equilibrium, within the resolution of the FS, without collapse of the transient. Below the critical intensity the flow rule hardly matters: with 18 mm/h the minimum FS is 1.316 with $\psi = 0$ and 1.319 with $\psi = \phi'/2$ and ϕ'. With $\delta = 0.01$ the critical intensity is 26 mm/h with $\psi = 0$ and 31 mm/h with $\psi = \phi'$ (20% higher); with $\psi = 0$ the failure occurs 5.4–6.3 h earlier in the rainfalls of 54 and 90 mm/h (Table 3). With $\psi = \phi'/2$ only 18 and 36 mm/h were computed below 54 mm/h, so its critical intensity is only known to lie between 18 and 36 mm/h.

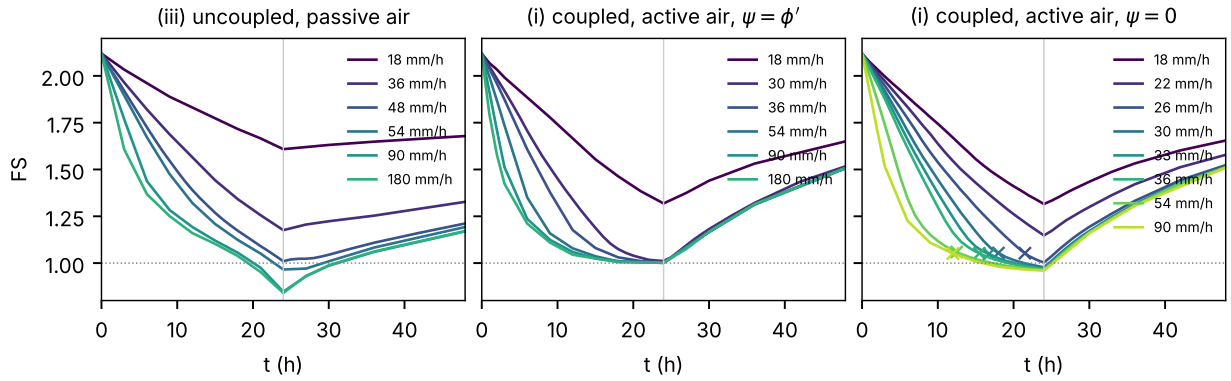


Figure 6: FS during and after 24-h rainfalls of several intensities, soil f_{50}, $\alpha = 45°$: model (iii) and model (i) with $\psi = \phi'$ and $\psi = 0$ (×: loss of equilibrium of the coupled transient; afterwards the transient continues with the strength increased, only to follow the pressures).

With $\psi = 0$ the loss of equilibrium occurs when the FS of the strength reduction, computed with $\psi = \phi'$, is still 1.047–1.059. Either the associated strength reduction overestimates by about 5% the FS of a soil without dilatancy, or part of the loss of equilibrium is numerical; the time step does not explain it: with a time step of 0.125 h instead of 0.25 h during the rainfall, the transient with 36 mm/h loses equilibrium after 15.6 h instead of 15.5 h. For $\psi = 0$ the time of failure is therefore t_f, and $22 < I_c \leq 26$ mm/h by the loss of equilibrium.

The FS values below one after t_f use the pressures of a transient continued with the strength increased and only show that the equilibrium was exceeded.

The flow rule matters as much with passive air. In model (ii) the transient without dilatancy loses equilibrium after 22.3 h with 54 mm/h, and its critical intensity is 53 mm/h; with $\psi = \phi'$ the minimum FS is 1.033, 1.014, 1.010 and 1.009 with 54, 66, 78 and 90 mm/h, and the transient never loses equilibrium. In both coupled models, the transients with an associated flow rule do not collapse in the range computed, and those without dilatancy do. The mechanism (for instance the suction generated by the plastic dilation of the shear zone) was not isolated here.

8. Generalization

Table 4 extends the comparison of models (i) and (iii) to the permeable f_{10} soil, to the Bishop parameter $\chi = S_e$ and to the low-permeability f_{100} soil on two slopes. The three soils of Rahardjo et al. (2007) differ both in k_s and in the retention parameter a (Table 2), so the effect of k_s alone is not isolated by these comparisons.

Table 4: Critical intensity $I_{c,\delta}$ (mm/h) of 24-h rainfalls in the usual model (iii) and in the complete model (i, $\psi = \phi'$), and the lowest minimum FS computed. $>$: no failure up to the most intense rainfall computed; $\leq$: upper bound (the run at the upper intensity stopped at the first FS ≤ 1.01, so its minimum FS is not known); n.c.: not computable with the runs available.

		model (iii)					model (i), $\psi = \phi'$				
slope and soil	k_s	0.02	0.01	0.005	0	min. FS	0.02	0.01	0.005	0	min. FS
f_{50}, 45°, $\chi = 1$ (reference)	36	47	48	49	49	0.841	30	31	35	> 180	1.000
f_{10}, 45°, $\chi = 1$	360	$\leq$129	$\leq$135	$\leq$138	n.c.	0.841	$\leq$129	$\leq$139	n.c.	n.c.	1.029
f_{50}, 45°, $\chi = S_e$	36	47	$\leq$48	$\leq$51	$\leq$53	1.010	$\leq$27	$\leq$30	n.c.	n.c.	1.034
f_{100}, 63.4°, $\chi = 1$	3.6	> 80	> 80	> 80	> 80	1.093	29	> 80	> 80	> 80	1.013
f_{100}, 45°, $H_s = 20$ m, $\chi = 1$	3.6	> 80	> 80	> 80	> 80	1.252	> 80	> 80	> 80	> 80	1.172

Permeable soil. In the f_{10} soil ($k_s = 360$ mm/h) the bias changes sign: the usual model gives $I_{c,\delta} = 125$, 128 and 129 mm/h for $\delta = 0.02$, 0.01 and 0.005, and the complete model 127, 134 and 138 mm/h, 2–6% more; with $\delta = 0$ the usual model gives 131 mm/h, while the minimum FS of the complete model with 140 mm/h is 1.0015. The critical intensities are 0.35–$0.38 k_s$. The surface also ponds in this soil: with 140 mm/h in model (i) the first vertex ponds after 6.75 h, 87 surface vertices are ponded when the FS reaches 1.01 (13 h) and the air pressure reaches 40 kPa. The different behaviour of this soil is therefore not explained by the absence of ponding; the contributions of the air and of the coupling were not separated for it.

Low-permeability soil. In the f_{100} soil ($k_s = 3.6$ mm/h) no 24-h rainfall of up to 80 mm/h brings a 20-m, 45° slope to failure in either model; the complete model gives a minimum FS 0.08–0.11 lower (1.172 against 1.252 with 80 mm/h), and its minimum occurs after the rainfall, 30–36 h after its start, whereas that of model (iii) occurs at 48 h, the end of the simulation. On the 63.4° slope model (iii) stays 9–10% above failure at any intensity, whereas model (i) comes within 0.013–0.015 of one (Section 6).

Bishop parameter. With $\chi = S_e$ (for the Fredlund–Xing curve with $C(\psi) = 1$, $S_e = S_w$) the suction contributes less to the initial strength (initial FS 1.895 instead of 2.119) and

the air pressure also enters the effective stress, through $(1 - \chi)p_a$. The critical intensity is 47–49 mm/h in model (iii) and 27–30 mm/h in model (i) ($\delta = 0.005$–0.02): the usual model overestimates it by 64–77%, more than with $\chi = 1$ (40–59%). With $\delta = 0$ model (iii) gives 50 mm/h; the minimum FS of model (i) with 30 mm/h is 1.003, and its run with 36 mm/h stopped at the first FS ≤ 1.01.

Confinement of the air. Opening the base and the lateral boundaries to the air changes the minimum FS by less than 0.002 (Section 6.1).

Mesh. With elements of 0.35 m instead of 0.75 m near the surface (7732 triangles instead of 1684; initial FS 2.106 instead of 2.119), the minimum FS of model (i) with 30 mm/h falls from 1.011 to 1.005 and that of model (iii) with 48 mm/h from 1.010 to 0.987 (Fig. 5). With the local slope of the minimum FS against the intensity, these differences shift the critical intensities $I_{c,0.01}$ to about 30 and 46 mm/h; their ratio, 1.5, does not change.

9. The results and the statements of the literature

Each item below quotes a statement of the literature and gives the corresponding result of this study.

1. *Atmospheric pore air.* Sheng et al. (2008, p. 811) consider the atmospheric air pressure "approximately true for most in situ conditions"; Fredlund (2006, p. 292) restricts it to soils "exposed to atmospheric pressure" with a continuous air phase. In our analyses the maximum air pressure reaches 13 kPa after 3 h of 9 mm/h on the f_{100} slope at 63.4°, before the first vertex ponds (6 h), and 36–38 kPa at the end of the 24-h rainfalls of 9–80 mm/h on the same slope, when 83 to all 96 surface vertices are ponded; in the f_{10} soil it reaches 40 kPa with 140 mm/h. With active air the critical intensity of the reference slope is about half of that with passive air (Section 5). Once the surface ponds, the soil below it is no longer exposed to the atmosphere, the situation excluded by Fredlund (2006).
2. *Is the passive-air assumption unrealistic?* Khoei and Saeedmonir (2021, p. 2) state that it leads "to unrealistic outcomes" and cite Callari and Abati (2009) among the works that adopt it; Callari and Abati (2009, p. 491) obtain "good agreement with available experimental data" in the Liakopoulos test with it. In our analyses both statements hold for different processes: in drainage (Liakopoulos) the passive-air solution reproduces the measurements and our two-phase solution drains more slowly than the test (Section 3); in infiltration with a ponded surface the active air changes the critical intensity by a factor of about two. Neither statement is general.
3. *Coupled or uncoupled.* Pradhan et al. (2022, p. 2) report that coupled analyses "have been reported to provide a better prediction of factor of safety"; Fredlund (2006, p. 315) that "the uncoupled simultaneous solution yields satisfactory results for geotechnical engineering purposes". In our analyses the coupling changes the critical intensity by 5–11% without dilatancy and by 27–62% with an associated flow rule ($\delta = 0.01$): the answer depends on the flow rule, which neither study discusses.
4. *Flow rule of the coupled analyses.* The coupled analyses of slope collapse of Wang et al. (2018, p. 6) and Liu and Wang (2021, p. 5) use dilation angles of 0.1° and 0°. In our coupled analyses the transient collapses without dilatancy ($I_r \geq 26\,\mathrm{mm/h}$) and does

not collapse with $\psi \geq \phi'/2$ (Section 7). Whether a coupled analysis predicts a collapse therefore depends on a parameter that these studies did not vary.

5. *The intensity that matters relative to k_s.* Liu and Wang (2021, p. 14), citing Rahardjo et al. (2007) among others, place the threshold of q/k_{sat} beyond which the FS no longer decreases at "about or slightly larger than 1". The threshold intensities of Rahardjo et al. (2007, p. 1539 and Fig. 7) are 140, 51 and 11 mm/h for the f_{10}, f_{50} and f_{100} soils, i.e. $0.39k_s$, $1.42k_s$ and $3k_s$, so the summary of Liu and Wang (2021) does not match the values of the study it cites. Our critical intensities are 0.35–$0.38k_s$ in the f_{10} soil in both models and $1.3k_s$ (model iii) and 0.7–$1.0k_s$ (model i) in the f_{50} soil: the ratio depends on the soil and on the model.
6. *The lower plateau.* The lower asymptotes of Rahardjo et al. (2007, Fig. 7) are 0.81 and 0.82 for the f_{10} and f_{50} soils and 1.45 for the f_{100} soil. Li et al. (2024, p. 9) find that the FS of a slope saturated by a steady infiltration "is identical to that without considering suction", and Cheng et al. (2024, p. 17) relate the constant minimum FS above a certain intensity to the degree of saturation, which "keeps constant". In our model (iii) the steady state with the surface ponded gives 0.84–0.85 for the three soils, 0.02–0.04 above the asymptotes of Rahardjo et al. (2007) for f_{10} and f_{50}; the f_{100} slope does not reach that state in 24 h, so its asymptote of 1.45 is not the FS of the saturated slope (Section 4).
7. *Low-permeability slopes.* Rahardjo et al. (2007, p. 1541): slopes with $k_s \leq 10^{-6}$ m/s "will not fail under short-duration rainfall ($T_r \leq 24$ h) even when the rainfall intensity is extremely high". Our model (iii) agrees (minimum FS 1.093 at 63.4°). With active air the minimum FS of the same slope is 1.013–1.015 (1.009 with the finest mesh): failure with $\delta = 0.02$, not with $\delta = 0.01$ (Section 6).
8. *The minimum FS as the measure of safety.* Tang et al. (2018, p. 324) adopt the minimum FS during and after the rainfall "to reflect the most unstable state of the slope". In our coupled models with an associated flow rule the minimum FS did not fall below one in the cases computed (Section 2.4), so beyond failure it does not distinguish the rainfalls; models with different coupling can only be compared by the rainfall that brings the FS to one.

10. Implications for rainfall thresholds

For the reference slope and 24-h rainfalls, the critical intensity of the usual model is $1.3k_s$ and that of the complete model 0.8–$1.0k_s$ with an associated flow rule and $0.7k_s$ without dilatancy ($\delta = 0.005$–0.02). A threshold derived from the usual chain is thus not conservative for this slope: rainfalls between the two thresholds, which the usual analysis classifies as safe, bring the complete model to failure. On an intensity–duration–frequency curve a lower intensity of the same duration has a shorter return period; by how much depends on the local curve, which is not part of this study. Rahardjo et al. (2007, p. 1541) conclude that low permeability prevents the failure under short rainfalls; for the f_{100} slope at 63.4° this holds in model (iii) but, in model (i), only with $\delta \leq 0.01$ (Section 6).

The bias is not uniform. It is large in the f_{50} soil and on the f_{100} slope at 63.4°, and reversed (2–6%) in the f_{10} soil, although the surface ponds and the air pressure rises in all three (Sections 6 and 8); the reason for the difference was not established here. A two-phase

flow analysis on a rigid skeleton (model iv) gives a critical intensity 5–6% below that of the complete model without dilatancy and 18–29% below that with an associated flow rule; the usual single-phase seepage analysis gives one 85–88% and 40–59% above them.

11. Limitations

- Two-dimensional, homogeneous slope with a single geometry for the decomposition; the critical intensity of other slopes was examined only for two soils, two Bishop laws and one steeper slope.

- The air leaves only through the ground surface, and once it ponds only through the minimum relative permeability $k_{ra}^{\min}$. Opening the base and the lateral boundaries does not change the results (Section 6.1); with $k_{ra}^{\min} = 10^{-2}$ the minimum FS of the f_{100} slope rises by 0.03. Macropores, cracks, vegetation and heterogeneities are represented only through $k_{ra}^{\min}$.

- The strength reduction is associated even when the transient uses $\psi = 0$; the loss of equilibrium of the transient is then the failure indicator, and part of it may be numerical.

- No hysteresis of the retention curve, constant porosity in the hydraulic coefficients, no dissolved air, no vapour and an isothermal ideal gas. The air compressibility is integrated with the saturation of the end of the step, which is not exactly mass conservative for the air (Section 2).

- The mesh error of the minimum FS was assessed in three cases (Fig. 5): with 0.35-m elements it falls by 0.006–0.023 and, in the f_{100} case, by about 0.003 per refinement without clear convergence; the critical intensities shift by 1–2 mm/h.

- The air phase was verified only through the mass balance and was not validated: in the Liakopoulos test the two-phase solution drains more slowly than the measurements, and no measurement of the air pressure under infiltration was used.

- The pressure profiles of Fig. 4c,d were computed from snapshots that are not part of the published data set; the other results are.

12. Conclusions

A coupled formulation with displacement, pore-water and pore-air pressures was implemented in NeoPZ, verified against the Liakopoulos drainage test with passive air and used to quantify how the hypotheses of the usual chain of analysis, seepage on a rigid skeleton with atmospheric pore air followed by a stability analysis with the computed pressures, change the rainfall that brings an unsaturated slope to failure. The conclusions refer to the slope of Rahardjo et al. (2007) and the cases computed.

1. In the coupled models with an associated flow rule the transient is solved with the real strength, and the minimum FS of the strength reduction did not fall below one in the cases computed (1.000–1.014 for $I_r \geq I_c$). The models must therefore be compared by

the critical intensity $I_{c,\delta}$ for a stated threshold $1+\delta$ and by the time to failure, not by the minimum FS; with $\delta = 0$ the complete associated model has no critical intensity up to 180 mm/h.

2. For $k_s = 36\,\mathrm{mm/h}$ and 24-h rainfalls the usual model (iii) gives $I_{c,\delta} = 47$–49 mm/h ($1.3k_s$) and the complete model (i) 30–35 mm/h with $\psi = \phi'$ and 26 mm/h with $\psi = 0$ ($\delta = 0.005$–0.02). The usual hypotheses overestimate the failure-triggering rainfall by 40–59% with $\psi = \phi'$ and by 85–88% with $\psi = 0$, and delay the predicted failure by 4.5 h ($\psi = \phi'$) and 9.9 h ($\psi = 0$) with 54 mm/h.
3. The two simplifications act in opposite directions. With the same coupling, active air lowers $I_{c,\delta}$ by a factor of 2.0–2.5 (from 48 to 25 mm/h on a rigid skeleton, $\delta = 0.01$); the coupling raises it, by 5–11% without dilatancy (at most 11% with passive air) and by 27–62% with $\psi = \phi'$ ($\delta = 0.01$). The air effect is the larger one, and a two-phase flow analysis on a rigid skeleton (model iv) gives a critical intensity below that of the complete model.
4. With active air, less water infiltrates and the matric suction below the face is higher than with passive air, but the pore-water pressure is also higher and the FS lower. On the low-permeability slope at 63.4° ($k_s = 3.6\,\mathrm{mm/h}$), which model (iii) keeps 9–10% above failure, the complete model gives a minimum FS of 1.013–1.015 (1.009 with the finest mesh). Opening the base and the lateral boundaries to the air does not change these results.
5. Without dilatancy the coupled transient loses equilibrium for $I_r \geq 26\,\mathrm{mm/h}$; with $\psi \geq \phi'/2$ it does not in the range computed. Whether a coupled analysis predicts a collapse therefore depends on the flow rule.
6. The bias persists with $\chi = S_e$ (64–77%) and with a finer mesh (about 46 against 30 mm/h). In the permeable f_{10} soil it changes sign: the usual model gives a critical intensity 2–6% below that of the complete model, although the surface ponds and the air pressure rises; the reason was not established.

Several statements of the literature are questioned or qualified by these results (Section 9): in our analyses the pore air is far from atmospheric once the surface ponds, whereas Sheng et al. (2008) consider the atmospheric value "approximately true for most in situ conditions"; neither the passive-air assumption nor the uncoupled solution is generally adequate or inadequate, the answer depending on the process and on the flow rule; and the threshold intensity relative to k_s is not "about or slightly larger than 1" in the study that Liu and Wang (2021) cite for it. The validation of the air phase against measured air pressures under infiltration, natural (open, heterogeneous) slopes and three-dimensional effects are the next steps.

Code and data availability

The material `TPZMatUnsaturatedUPwPa`, the program `SlopeRainfall` and the scripts that run all the cases and draw the figures are in the repository `https://github.com/diogocecilio/neopz-master`, a fork of the NeoPZ library; the results are in the folder `Projects/SlopeRainfall/resultados` of that repository.

References

Bergamaschi, L., Putti, M., 1999. Mixed finite elements and Newton-type linearizations for the solution of Richards' equation. International Journal for Numerical Methods in Engineering 45, 1025–1046.

Bishop, A.W., 1959. The principle of effective stress. Teknisk Ukeblad 106(39), 859–863.

Caine, N., 1980. The rainfall intensity–duration control of shallow landslides and debris flows. Geografiska Annaler 62(1–2), 23–27.

Callari, C., Abati, A., 2009. Finite element methods for unsaturated porous solids and their application to dam engineering problems. Computers & Structures 87, 485–501.

Cheng, H., Wu, Z., Chen, H., Zhou, X., 2024. Stability analysis of unsaturated–saturated soil slopes under rainfall infiltration using the rigorous limit equilibrium method. Bulletin of Engineering Geology and the Environment 83, 147. https://doi.org/10.1007/s10064-024-03623-y

Devloo, P.R.B., 1997. PZ: an object oriented environment for scientific programming. Computer Methods in Applied Mechanics and Engineering 150, 133–153. https://doi.org/10.1016/s0045-7825(97)00097-2

Fredlund, D.G., 2006. Unsaturated soil mechanics in engineering practice. Journal of Geotechnical and Geoenvironmental Engineering 132, 286–321.

Fredlund, D.G., Morgenstern, N.R., Widger, R.A., 1978. The shear strength of unsaturated soils. Canadian Geotechnical Journal 15(3), 313–321.

Fredlund, D.G., Xing, A., 1994. Equations for the soil-water characteristic curve. Canadian Geotechnical Journal 31, 521–532. https://doi.org/10.1139/t94-061

Ghorbani, J., Airey, D.W., El-Zein, A., 2018. Numerical framework for considering the dependency of SWCCs on volume changes and their hysteretic responses in modelling elastoplastic response of unsaturated soils. Computer Methods in Applied Mechanics and Engineering (accepted manuscript). https://doi.org/10.1016/j.cma.2018.03.008

Griffiths, D.V., Lane, P.A., 1999. Slope stability analysis by finite elements. Géotechnique 49(3), 387–403.

Guzzetti, F., Peruccacci, S., Rossi, M., Stark, C.P., 2008. The rainfall intensity–duration control of shallow landslides and debris flows: an update. Landslides 5(1), 3–17.

Khoei, A.R., Saeedmonir, S., 2021. Computational homogenization of fully coupled multiphase flow in deformable porous media. Computer Methods in Applied Mechanics and Engineering 376, 113660.

Leong, E.C., Rahardjo, H., 1997. Permeability functions for unsaturated soils. Journal of Geotechnical and Geoenvironmental Engineering 123(12), 1118–1126.

Li, X., Thomas, H.R., Fan, Y., 1999. Finite element method and constitutive modelling and computation for unsaturated soils. Computer Methods in Applied Mechanics and Engineering 169, 135–159.

Li, Y., Tao, X., Xu, J., Hou, C., Sun, Z., 2024. Unsaturated slope stability analysis for cohesive soil with tensile strength cut-off. Engineering Geology 339, 107639.

Liakopoulos, A.C., 1965. Transient flow through unsaturated porous media. PhD thesis, University of California.

Liu, X., Wang, Y., 2021. Probabilistic simulation of entire process of rainfall-induced landslides using random finite element and material point methods with hydro-mechanical coupling. Computers and Geotechnics 132, 103989.

Pedroso, D.M., 2015. A solution to transient seepage in unsaturated porous media. Computer Methods in Applied Mechanics and Engineering 285, 791–816. https://doi.org/10.1016/j.cma.2014.12.009

Pradhan, S., Toll, D.G., Rosser, N.J., Brain, M.J., 2022. An investigation of the combined effect of rainfall and road cut on landsliding. Engineering Geology 307, 106787. https://doi.org/10.1016/j.enggeo.2022.106787

Rahardjo, H., Ong, T.H., Rezaur, R.B., Leong, E.C., 2007. Factors controlling instability of homogeneous soil slopes under rainfall. Journal of Geotechnical and Geoenvironmental Engineering 133(12), 1532–1543.

Sheng, D., Gens, A., Fredlund, D.G., Sloan, S.W., 2008. Unsaturated soils: from constitutive modelling to numerical algorithms. Computers and Geotechnics 35, 810–824. https://doi.org/10.1016/j.compgeo.2008.08.011

Tang, G., Huang, J., Sheng, D., Sloan, S.W., 2018. Stability analysis of unsaturated soil slopes under random rainfall patterns. Engineering Geology 245, 322–332.

Wang, B., Vardon, P.J., Hicks, M.A., 2018. Rainfall-induced slope collapse with coupled material point method. Engineering Geology 239, 1–12.

Wang, S., Idinger, G., Wu, W., 2021. Centrifuge modelling of rainfall-induced slope failure in variably saturated soil. Acta Geotechnica 16, 2899–2916. https://doi.org/10.1007/s11440-021-01169-x

Zhang, L.L., Fredlund, D.G., Fredlund, M.D., Wilson, G.W., 2014. Modeling the unsaturated soil zone in slope stability analysis. Canadian Geotechnical Journal 51, 1384–1398.